Kernix 0.11.0 / Performance brief

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

34 workloads

120 observations

Source precision retained

Matrix multiplication

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

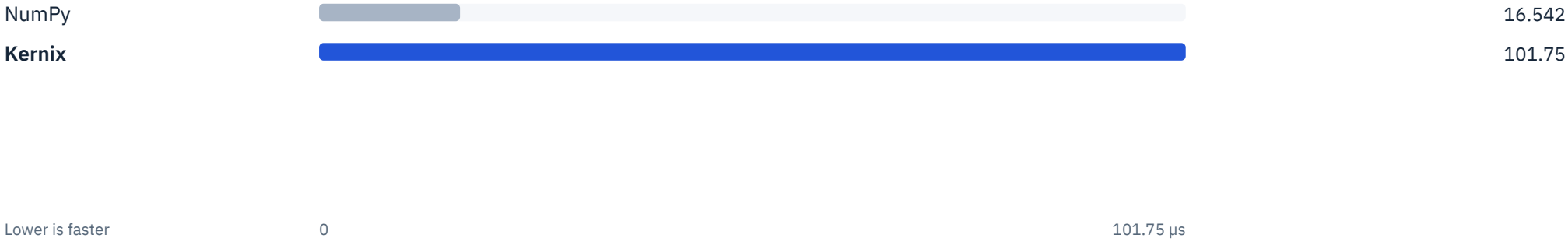
Kernix 0.11.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Independent matrix operands

Per-call median

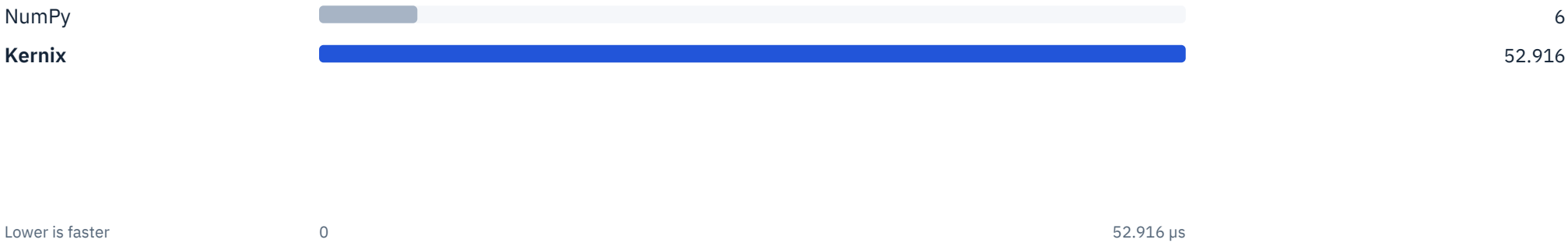
128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · independent operands



Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · independent operands



Kernix 0.11.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matrix multiplication

Per-call median

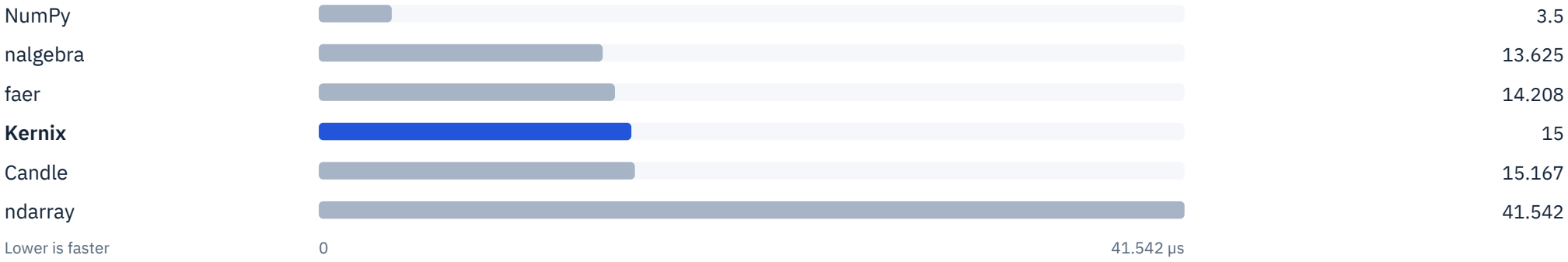
32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



Kernix 0.11.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



Kernix 0.11.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matrix multiplication

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product

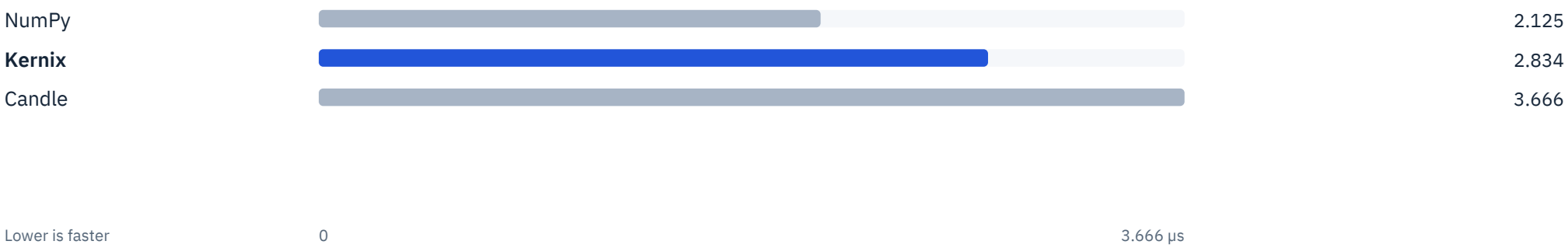


Kernix 0.11.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



Kernix 0.11.0 / Explicit parallel execution

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matmul · 2 workers · reused output

Per-call median

64 × 64 · 64 × 64 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f64 · 2 workers · reused output · pool preparation excluded



Kernix 0.11.0 / Explicit parallel execution

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matmul · 2 workers · reused output

Per-call median

256 × 256 · 256 × 256 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f32 · 2 workers · reused output · pool preparation excluded



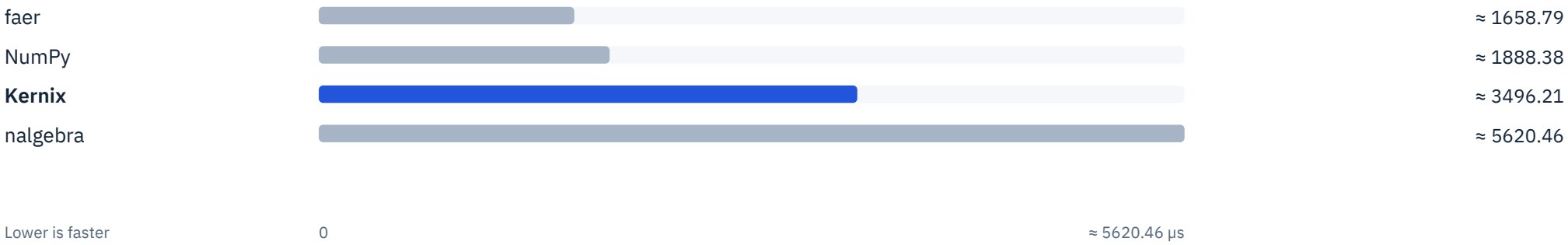
Kernix 0.11.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Cholesky factorization

512 × 512 · f64 · 1 worker · allocated output

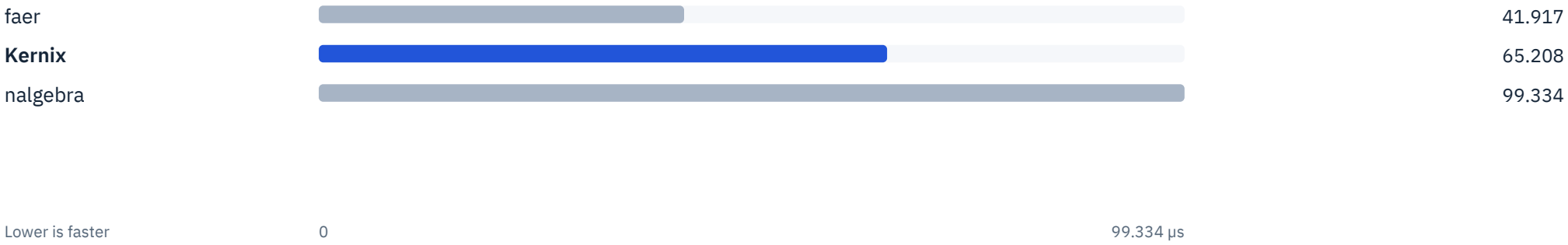
Per-call median



Column-pivoted QR

96 × 32 · f64 · 1 worker · allocated output

Per-call median



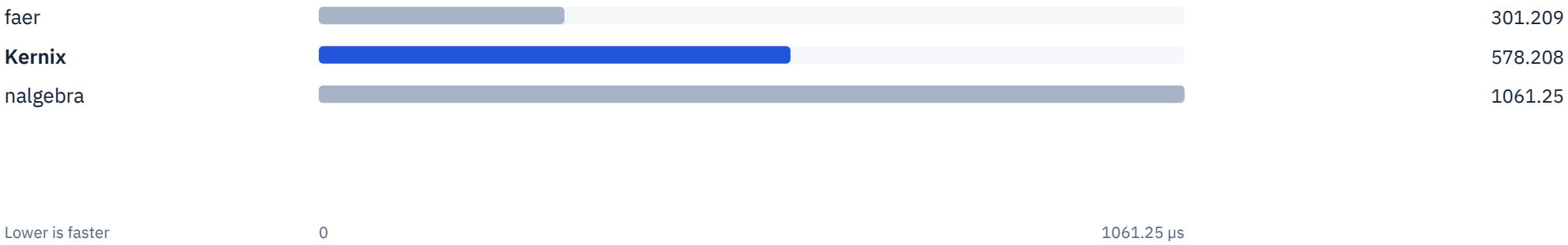
Kernix 0.11.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Column-pivoted QR

256 × 64 · f64 · 1 worker · allocated output

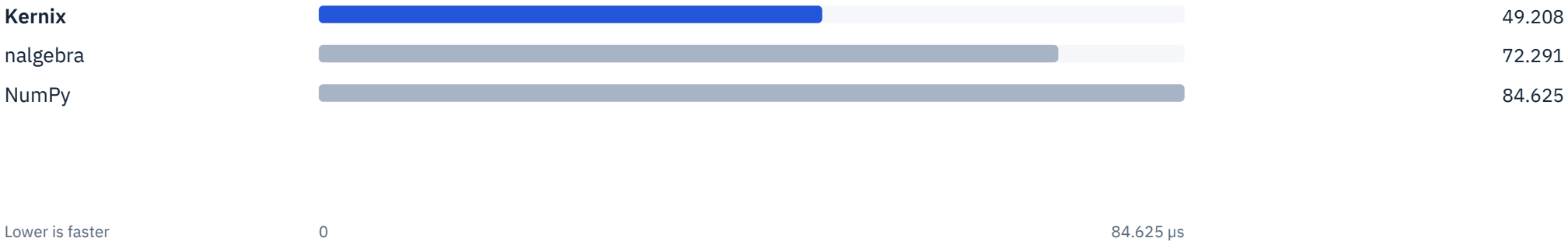
Per-call median



Minimum-norm least squares

64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ

Per-call median



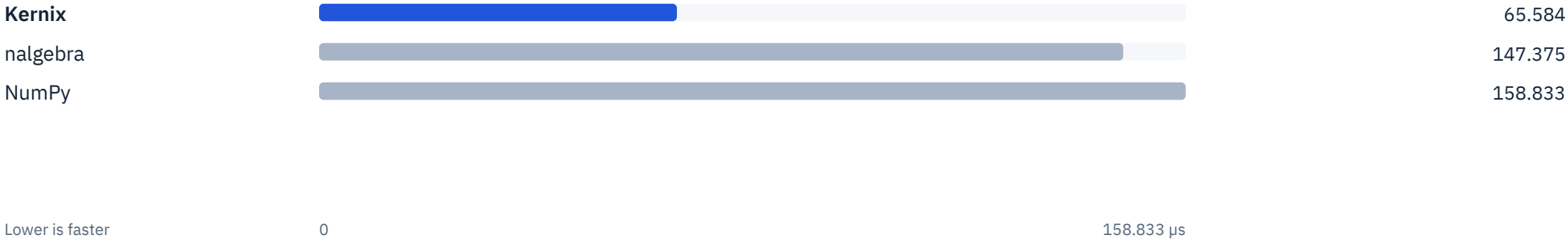
Kernix 0.11.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Minimum-norm least squares

Per-call median

32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Thin QR

Per-call median

96 × 32 · f64 · 1 worker · allocated output



Kernix 0.11.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Thin QR

256 × 64 · f64 · 1 worker · allocated output



Cholesky factorization

64 × 64 · f64 · 1 worker · allocated output



Kernix 0.11.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Cholesky factorization

Per-call median

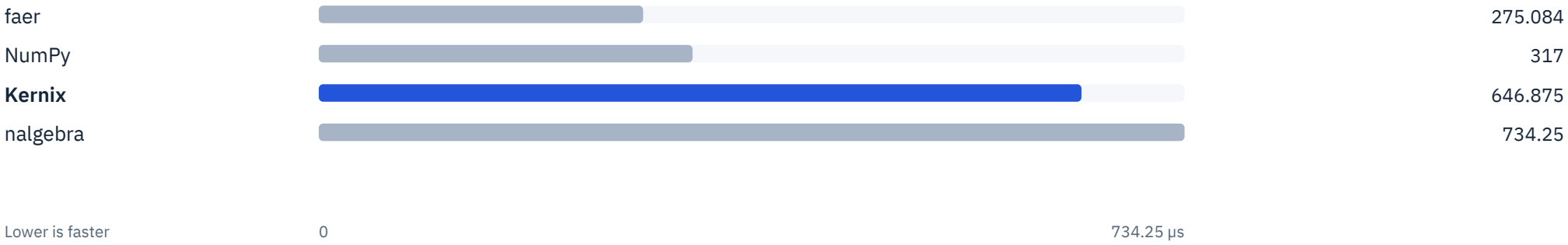
128 × 128 · f64 · 1 worker · allocated output



Cholesky factorization

Per-call median

256 × 256 · f64 · 1 worker · allocated output



Kernix 0.11.0 / Training & autodiff

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Shared-variable backward

Per-call median

sum(X * X), 64 × 32; prepared graph · f64 · 1 worker · prepared graph · fresh gradients



Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Kernix 0.11.0 / Training & autodiff

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Model inference

Per-call median

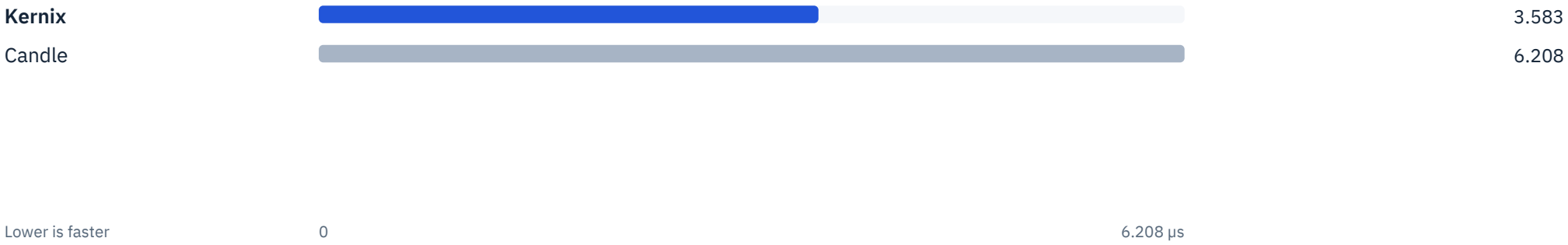
batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model recording + forward

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.11.0 / Training & autodiff

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented

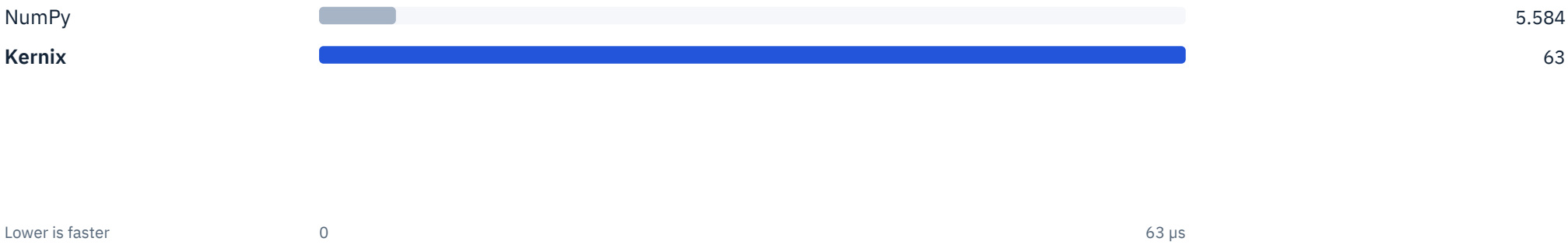


Kernix 0.11.0 / Tensor contractions

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Coalesced tensor contraction

16 × 32 × 8 · 32 × 8 × 16 · f64 · 1 worker · allocated output



Kernix 0.11.0 / Statistics & activations

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Row softmax

256 × 64 · f64 · 1 worker · allocated output



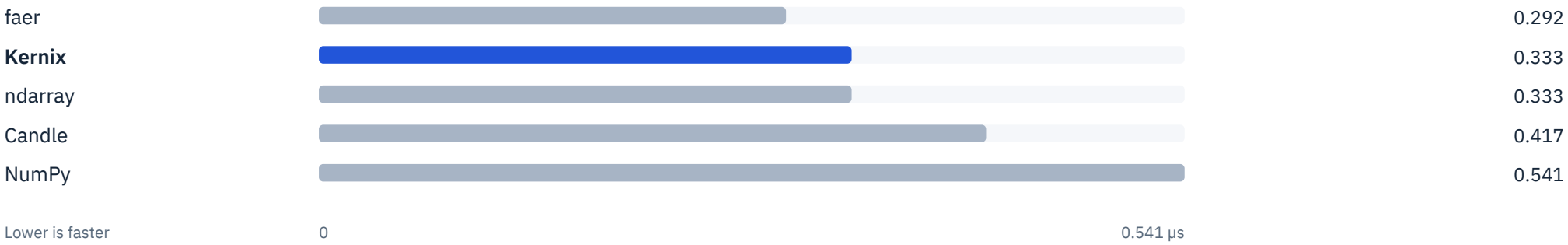
Kernix 0.11.0 / Tensor construction

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Create zeros

64 × 64 · f32 · 1 worker · allocated output

Per-call median



Create ones

64 × 64 · f32 · 1 worker · allocated output

Per-call median



Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9811495292521482	0.9640939796747162 – 1.0317623762376238

Matrix multiplication

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0179028132992327	0.9799530431641683 – 1.057152803374698

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

256 × 256 · 256 × 256 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0055372096965587	0.9167619146578 – 1.0116898335997095

Matrix multiplication

256 × 256 · 256 × 256 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9812478690760313	0.9126053787548315 – 1.0005781117979278

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Independent matrix operands

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	0.9812	95% INTERVAL	0.9602777248236628 – 1.0004283818658641

Independent matrix operands

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	1	95% INTERVAL	0.9825582554268742 – 1.0363957812595257

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0010035092779541	0.9997475582268971 – 1.0615615615615615

Cholesky factorization

256 × 256 · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.520454357259916	0.5096753203342619 – 0.521812129502964

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

512 × 512 · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.3718924151845818	0.3676295604130526 – 0.39798777381558836

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.9097270142703359	0.8939063004846527 – 0.9683136560953355

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Matmul · 2 workers · reused output

256 × 256 · 256 × 256 · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.8891034482758621	0.8764368128463624 – 0.9266004415011038

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f32

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.7962008346079043	0.7679718133772803 – 0.8294814814814815

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f64	16542	16.542
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f64	101750	101.75
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f32	6000	6
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f32	52916	52.916
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	2083	2.083
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	2125	2.125
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	2542	2.542
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	1375	1.375
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	2458	2.458
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	2291	2.291

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	15167	15.167
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	41542	41.542
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	15000	15
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	13625	13.625
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	3500	3.5
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	14208	14.208
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	108542	108.542
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	100917	100.917
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	95125	95.125
Matrix multiplication / Kernix · explicit Accelerate 128 × 128 · 128 × 128 · f64	16167	16.167

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	110583	110.583
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	111833	111.833
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	17291	17.291
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	792250	792.25
Matrix multiplication / Kernix · explicit Accelerate 256 × 256 · 256 × 256 · f64	101958	101.958
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	753166	753.166
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	753208	753.208
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	871209	871.209
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	115875	115.875
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	804417	804.417

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	7166	7.166
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	7584	7.584
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	8833	8.833
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	6875	6.875
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	1916	1.916
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	22250	22.25
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	48875	48.875
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	57292	57.292
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	54000	54
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	5250	5.25

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	55500	55.5
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	51333	51.333
Matrix multiplication / Kernix · explicit Accelerate 128 × 128 · 128 × 128 · f32	6500	6.5
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	366917	366.917
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	35834	35.834
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	424042	424.042
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	436833	436.833
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	398125	398.125
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	400541	400.541
Matrix multiplication / Kernix · explicit Accelerate 256 × 256 · 256 × 256 · f32	42292	42.292

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	3666	3.666
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	2125	2.125
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	2834	2.834
Matmul · 2 workers · reused output / Kernix 64 × 64 · 64 × 64 · f64	22625	22.625
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f64	65666	65.666
Matmul · 2 workers · reused output / Kernix 256 × 256 · 256 × 256 · f64	447583	447.583
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f32	55291	55.291
Cholesky factorization / nalgebra 512 × 512 · f64	5620458	5620.458
Cholesky factorization / NumPy 512 × 512 · f64	1888375	1888.375
Cholesky factorization / faer 512 × 512 · f64	1658792	1658.792

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / Kernix 512 × 512 · f64	3496208	3496.208
Column-pivoted QR / nalgebra 96 × 32 · f64	99334	99.334
Column-pivoted QR / Kernix 96 × 32 · f64	65208	65.208
Column-pivoted QR / faer 96 × 32 · f64	41917	41.917
Column-pivoted QR / Kernix 256 × 64 · f64	578208	578.208
Column-pivoted QR / nalgebra 256 × 64 · f64	1061250	1061.25
Column-pivoted QR / faer 256 × 64 · f64	301209	301.209
Minimum-norm least squares / NumPy 64 × 32, rank 16, 4 RHS · f64	84625	84.625
Minimum-norm least squares / Kernix 64 × 32, rank 16, 4 RHS · f64	49208	49.208
Minimum-norm least squares / nalgebra 64 × 32, rank 16, 4 RHS · f64	72291	72.291

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Minimum-norm least squares / nalgebra 32 × 64, rank 32, 4 RHS · f64	147375	147.375
Minimum-norm least squares / NumPy 32 × 64, rank 32, 4 RHS · f64	158833	158.833
Minimum-norm least squares / Kernix 32 × 64, rank 32, 4 RHS · f64	65584	65.584
Thin QR / faer 96 × 32 · f64	52416	52.416
Thin QR / NumPy 96 × 32 · f64	48959	48.959
Thin QR / nalgebra 96 × 32 · f64	44000	44
Thin QR / Kernix 96 × 32 · f64	61958.00000000001	61.95800000000001
Thin QR / faer 256 × 64 · f64	326667	326.667
Thin QR / nalgebra 256 × 64 · f64	328375	328.375
Thin QR / Kernix 256 × 64 · f64	577750	577.75

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / NumPy 256 × 64 · f64	401708	401.708
Cholesky factorization / Kernix 64 × 64 · f64	22708	22.708
Cholesky factorization / faer 64 × 64 · f64	10292	10.292
Cholesky factorization / nalgebra 64 × 64 · f64	21084	21.084
Cholesky factorization / NumPy 64 × 64 · f64	16042	16.042
Cholesky factorization / Kernix 128 × 128 · f64	166459	166.459
Cholesky factorization / nalgebra 128 × 128 · f64	113792	113.792
Cholesky factorization / faer 128 × 128 · f64	47083	47.083
Cholesky factorization / NumPy 128 × 128 · f64	51250	51.25
Cholesky factorization / NumPy 256 × 256 · f64	317000	317

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / Kernix 256 × 256 · f64	646875	646.875
Cholesky factorization / faer 256 × 256 · f64	275084	275.084
Cholesky factorization / nalgebra 256 × 256 · f64	734250	734.25
Shared-variable backward / Candle sum(X * X), 64 × 32; prepared graph · f64	4250	4.25
Shared-variable backward / Kernix sum(X * X), 64 × 32; prepared graph · f64	6125	6.125
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	6541	6.541
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	3250	3.25
Model inference / Candle batched normalized/sliced MLP · f64	5958	5.958
Model inference / Kernix batched normalized/sliced MLP · f64	2458	2.458
Model recording + forward / Kernix batched normalized/sliced MLP · f64	3583	3.583

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Model recording + forward / Candle batched normalized/sliced MLP · f64	6208	6.208
Model backward / Kernix prepared graph → fresh gradients · f64	3250	3.25
Model backward / Candle prepared graph → fresh gradients · f64	20958	20.958
Parameter update / Kernix transactional SGD, fixed gradients · f64	1875	1.875
Coalesced tensor contraction / Kernix 16 × 32 × 8 · 32 × 8 × 16 · f64	63000	63
Coalesced tensor contraction / NumPy 16 × 32 × 8 · 32 × 8 × 16 · f64	5584	5.584
Row softmax / ndarray 256 × 64 · f64	58875	58.875
Row softmax / Candle 256 × 64 · f64	66167	66.167
Row softmax / Kernix 256 × 64 · f64	76292	76.292
Row softmax / NumPy 256 × 64 · f64	83041	83.041

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / Kernix 64 × 64 · f32	333	0.333
Create zeros / faer 64 × 64 · f32	292	0.292
Create zeros / ndarray 64 × 64 · f32	333	0.333
Create zeros / Candle 64 × 64 · f32	417	0.417
Create zeros / NumPy 64 × 64 · f32	541	0.541
Create ones / NumPy 64 × 64 · f32	1209	1.209
Create ones / Candle 64 × 64 · f32	416	0.416
Create ones / Kernix 64 × 64 · f32	250	0.25
Create ones / ndarray 64 × 64 · f32	292	0.292
Create ones / faer 64 × 64 · f32	334	0.334